SOLUTIONS - Exercise - Week 5: MATH1061/7861

Problems. Prove or disprove the following statements.

(1) **(4 marks)** For all integers m and n, if $m^2 + n^2 + mn$ is even, then m and n are both even.

The statement is true. We will prove this by contraposition (contradiction also works).

Proof. The contrapositive is: For all integers m and n, if m is odd or n is odd, then $m^2 + n^2 + mn$ is odd.

Suppose that m is odd or n is odd. Without loss of generality, let m be odd. Then $m = 2k + 1$ for some $k \in \mathbb{Z}$.

If n is odd also, then $n = 2\ell + 1$ for some $\ell \in \mathbb{Z}$. Thus,

$$\begin{aligned} m^2 + n^2 + mn &= (2k+1)^2 + (2\ell+1)^2 + (2k+1)(2\ell+1) \\ &= 4k^2 + 4k + 1 + 4\ell^2 + 4\ell + 1 + 4k\ell + 2k + 2\ell + 1 \\ &= 2(2k^2 + 3k + 2\ell^2 + 3\ell + 2k\ell + 1) + 1 \end{aligned}$$

Since $k, \ell \in \mathbb{Z}$, $(2k^2 + 3k + 2\ell^2 + 3\ell + 2k\ell + 1) \in \mathbb{Z}$ and so $m^2 + n^2 + mn$ is odd.

If n is even instead, then $n = 2\ell$ for some $\ell \in \mathbb{Z}$ and

$$\begin{aligned} m^2 + n^2 + mn &= (2k+1)^2 + (2\ell)^2 + (2k+1)(2\ell) \\ &= 4k^2 + 4k + 1 + 4\ell^2 + 4k\ell + 2\ell \\ &= 2(2k^2 + 2k + 2\ell^2 + \ell + 2k\ell) + 1 \end{aligned}$$

Since $k, \ell \in \mathbb{Z}$, $(2k^2 + 2k + 2\ell^2 + \ell + 2k\ell) \in \mathbb{Z}$ and so $m^2 + n^2 + mn$ is odd. □

MARKING:
1 mark for mentioning that the statement is true.
1 mark for setting up the proof correctly (e.g. "The contrapositive is ..." or "Suppose, for the sake of contradiction, that ..." or "We will use proof by contraposition. Suppose that m is odd or n is odd..." or something similar).
2 marks for a valid proof of the statement. No marks for the proof if only an example for specific m and n is given.

(2) **(3 marks)** $\forall m, a, b \in \mathbb{Z}_{>0}$, $m \mid (ab)$ if and only if $m \mid a$ and $m \mid b$.

The statement is false. For a counterexample, consider $m = 6$, $a = 2$ and $b = 3$. Then $ab = 6$ so $m \mid (ab)$. But 6 does not divide 2 (nor 3).

Note that the backwards direction (if $m \mid a$ and $m \mid b$ then $m \mid (ab)$) is true, but students don't need to mention or prove this.

MARKING:
1 mark for mentioning that the statement is false.
2 marks for providing a valid counter-example.

(3) **(3 marks)** There exists an integer x such that $x^2 \equiv 3 \mod 5$.

Observe that x is congruent modulo 5 to a value in $\{0, 1, 2, 3, 4\}$.
If $x \equiv 0 \pmod 5$ then $x^2 \equiv 0 \cdot 0 \equiv 0 \pmod 5$.
If $x \equiv 1 \pmod 5$ then $x^2 \equiv 1 \cdot 1 \equiv 1 \pmod 5$.
If $x \equiv 2 \pmod 5$ then $x^2 \equiv 2 \cdot 2 \equiv 4 \pmod 5$.
If $x \equiv 3 \pmod 5$ then $x^2 \equiv 3 \cdot 3 \equiv 9 \equiv 4 \pmod 5$.
If $x \equiv 4 \pmod 5$ then $x^2 \equiv 4 \cdot 4 \equiv 16 \equiv 1 \pmod 5$.
Since in every case $x^2 \not\equiv 3$, there is no integer x such that $x^2 \equiv 3 \pmod 5$.

MARKING:
1 mark for mentioning that the statement is false.
2 marks for a valid justification. Remove 1 mark for minor mistakes in modular arithmetic.